



U.S. SECURITIES AND EXCHANGE COMMISSION

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The sample includes every order between 9:35 am and 4:00 pm. Amex, Arca, Nasdaq and NYSE conduct some form of open and/or closing auction. Most permit a mixture of normal orders with open- or close- specific orders. NYSE and Amex print the open with a special code. Frequently a series of trades and cancellations are printed immediately thereafter and these orders are often at different prices. Nasdaq stocks frequently open within a few hundred milliseconds of 9:30 am, but always. Filtering the first five minutes is a conservative effort to avoid some of the idiosyncrasies associated with the open. The summary results could be slightly biased only to the extent that orders that occur before 9:35 are traded or canceled after. A similar caveat applies to the close. Some exchanges permit regular orders to participate in the close along with close- specific orders. To the extent that these orders are traded in the close, a slight bias could occur. We also filter trades flagged as "Special" and trades associated with intra-day crosses or IPO halts.

The list of included stocks and exchange-traded products is sourced from the Center for Research in Security Prices (CRSP) daily stock database (SHRCD = 10, 11 for stocks; SHRCD = 73 for ETPs). The list is updated monthly with a 3 – 4 week lag from the final trading day of the previous month. We construct the following variables:

- **Cancel to Trade Ratio:** $\# \text{ of cancels} \div \# \text{ of trades}$
- **Trade to Order Volume:** $\text{trade volume} \div \text{order volume}$
- **Hidden Rate:** $\# \text{ of hidden trades} \div \# \text{ of trades}$
- **Hidden Volume:** $\text{hidden trade volume} \div \text{total trade volume}$
- **Odd Lot Rate:** $\# \text{ of odd lot trades} \div \# \text{ of trades}$
- **Odd Lot Volume:** $\text{odd lot volume} \div \text{total trade volume}$
- **Price** is the closing price for the day according to the CRSP database. When there is no closing trade, a quote midpoint is used in the absence of a close price.
- **Market Capitalization (Size):** $\text{Price} * \text{shares outstanding}$. Shares outstanding are sourced from the daily CRSP files.
- **Turnover:** $\# \text{ of shares traded} \div \# \text{ of shares outstanding}$
- **Volatility** is the daily standard deviation of 1 - minute quote midpoint returns

We use the term “Hidden Trades” to mean trade executions on an exchange that are the result of marketable order matching an undisplaced, or hidden, resting order. The exchange feeds do not generally provide messages related to the addition of undisplaced orders. But many provide an indicator or other method for identifying when trade executions (which are not themselves hidden) involved a resting undisplaced order. NYSE Ultra feed and Amex do not report trades against undisplaced orders on their direct level book feeds. However, January 2014 marks the first full month in which odd lot trades are reported to the SIP, thereby permitting a full reconciliation of trades on the direct feed with trades reported to the SIP. For NYSE and Amex, trades reported to the SIP but not reported on the direct level book feed are trades against undisplaced orders. Beginning May 1, 2017, the MIDAS switched from the NYSE Ultrabook level feed to the NYSE Integrated Feed (NIF). This feed provides an indicator for identifying trades involving resting undisplaced orders.

For each stock and ETP in the CRSP universe, order volume, trade volume, cancel counts, trade counts and some identifying information (hidden, odd lot) are cumulated over a daily interval. The Cancel to Trade ratio includes full and partial cancellations and trades. Partial executions are more likely than partial cancellations. The Cancel to Trade ratio reflects this disparity and is exactly what appears to be: the ratio of the number of all cancel events to the number of all trade events. However, it does not provide an exact numerical representation of what actually happens to orders. A round lot is defined as 100 shares for all but three tickers. BRK.A and SEB have a round lot of one. BH has a round lot of 10. All trades involving nine or fewer shares are considered odd lot trades.

Monthly percentile sorted (decile and quartile) results are constructed by summing the daily order volumes, trade volumes, cancellations, trades and other information within a particular decile or quartile and then computing the ratios. In other words, the ratios are unweighted (i.e., they are not equal-weighted within the decile or quartile). These results represent the expected outcome for a randomly selected order from within a particular decile or quartile of securities. Exchange-sorted results are constructed in a similar manner. Stocks and ETPs could be in different deciles or quartiles on different days. Data for the time series figures are constructed by summing the daily order volumes, trade volumes, cancels, trades and other information over the day and then computing the ratios.

The Monthly User Download Files contain each variable for each ticker, day and exchange. Due to the lower granularity of data reported using the level-book method compared to the order-based method (see [Data Highlight 2014-03, “Order Book Reporting Schemes and Their Impact on Some Market Activity Measures”](#)), and to avoid potential misinterpretations due to comparing such data across these mechanisms, the following metrics for NYSE and Amex will be removed from the website charts: Cancel-to-Trade Ratio, Odd Lot Ratio and Odd Lot Volume. Furthermore, the aggregate charts including the decile charts, for these three metrics will currently exclude the contributions from NYSE and Amex; the Quarterly User Files will also not include NYSE and Amex for these three measures. Beginning May 1, 2017, the Cancel-to-Trade Ratio, Odd Lot Ratio and Odd Lot Volume website charts will include data from the NYSE Integrated Feed, as will the aggregate charts and the Quarterly User Files.